📘 What is NLTK?

NLTK (Natural Language Toolkit) is a **Python library** used for working with **human language data (text)**.

👉 In simple words:

NLTK helps you **process, analyze, and understand text** using Python.

🔹 Full Form

- **NLTK = Natural Language Toolkit**

🔹 What can NLTK do?

NLTK provides tools for common **NLP (Natural Language Processing)** tasks:

✅ 1. Tokenization

- Breaking text into words or sentences

"I love AI" → ["I", "love", "AI"]

✅ 2. Stopword Removal

- Removing common words like: *is, the, and*

✅ 3. Stemming & Lemmatization

- Convert words to root form

running → run
better → good

✅ 4. Part-of-Speech (POS) Tagging

- Identify grammar roles

"I love AI"
→ I (pronoun), love (verb), AI (noun)

✅ 5. N-Grams

- Generate unigram, bigram, trigram

"I love AI"
→ ("I love"), ("love AI")

✅ **6. Text Classification**

- Spam detection, sentiment analysis, etc.

✅ **7. Corpora (Datasets)**

- Comes with built-in datasets like:
  - Brown Corpus
  - Gutenberg Corpus

🔹 **Example Code (Basic)**

```
import nltk
from nltk.tokenize import word_tokenize

text = "I love AI"
tokens = word_tokenize(text)

print(tokens)
```

Output:

```
['I', 'love', 'AI']
```

🔹 **Why use NLTK?**

- Beginner-friendly
- Easy to learn NLP basics
- Lots of built-in tools & datasets
- Great for **learning and experiments**

🔹 **Limitation**

- Not very fast for production
- Less powerful than modern libraries like:
  - spaCy
  - Transformers (Hugging Face)